Data Source - *NYC Trip Data*
We will analyse **2025** year data -> **Green Taxi trip** records only.

Project Steps
Step 1 – Download data from the above website
Green Taxi Trip Records (2026) in Parquet format – We will retrieve the data directly using ADF **(HTTP connection)**
Download **TaxiZone Lookup table** (csv format).

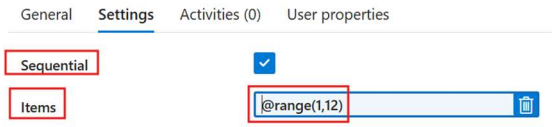
Step 2 – Set up MS Azure (BRONZE LAYER)
i. Create a **resource group, storage account** (data lake), 3 **containers** inside the storage account – *gold, silver, bronze*
ii. In the bronze container, create a directory (**Add Directory** option) – *trip_zone*. Upload the **TaxiZone Lookup table** csv file in this folder.
iii. Create **ADF resource**
iv. Create **Databricks resource**
v. Create **Microsoft Entra Id (service principal)** to give access to Databricks to Data lake

Step 3 – Create a pipeline in ADF (*to ingest data from the **NYC link** to the Bronze container*)
The site has data in **Parquet** format separated by each month.. i.e each month has different links –
Jan - https://d37ci6vzurychx.cloudfront.net/trip-data/green_tripdata_2025-**01**.parquet
Dec - https://d37ci6vzurychx.cloudfront.net/trip-data/green_tripdata_2025-**12**.parquet

Base url – highlighted in blue
Relative url – highlighted in yellow

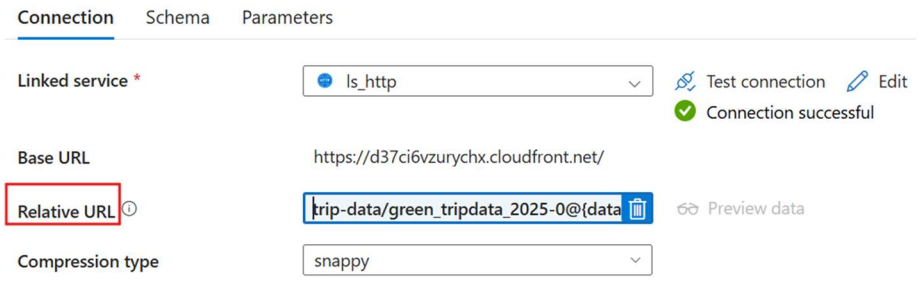
We will use **For Each loop** activity which will return 1, 2, 3, 4... as the output.

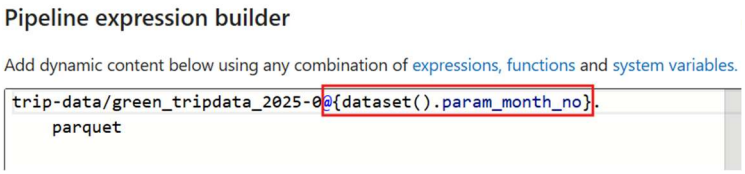
i. Configure ForEach loop activity



ii. Configure Copy activity
-> Add Copy activity inside ForEach **(Activities)**
Source dataset – HTTP, Parquet format
Source ls – Update the base url

Configuring parametrized relative url
-> Add dynamic content -> Create a parameter *param_month_no* -> Update **exp** as below –
`trip-data/green_tripdata_2025-0@{dataset().param_month_no}.parquet`

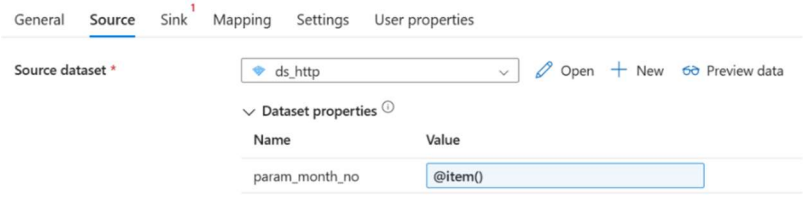




-> In the above exp, we are dynamically replacing the month no with the parameter that we have created.

Original rel url was - **`trip-data/green_tripdata_2025-01.parquet`**. So, we are replacing number 1 with the parameter created.

-> **Parameter value** will be the #time ForEach loop is running i.e update the exp as **`@item()`**.
Sink dataset – Azure Data Lake, Parquet format
Sink ls – Update the base url



-> When you run the above pipeline, it succeeds for months 1-9. However, it fails for 10-12.

-> To handle this, we need to add If Condition activity which will check the month no. and run the related copy activity accordingly.

ii. Configure If-Condition activity
-> Add **If-Condition activity** inside ForEach loop. Update **exp** to check if ForEach iteration is greater than 9 – **`@greater(item(),9)`**

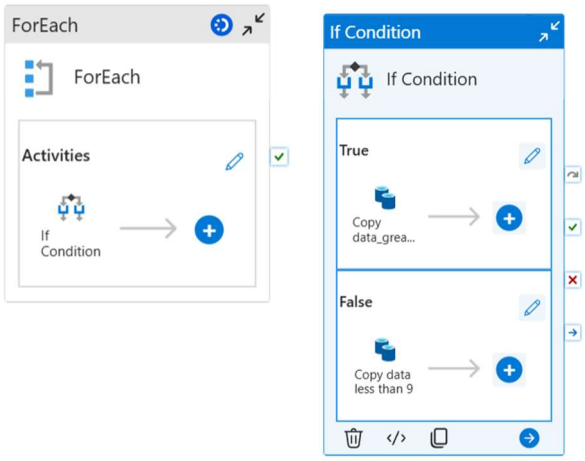
-> Copy the Copy activity that we configured earlier under **False** condition
-> Again, copy the same **Copy activity** under **True** condition. However, we need to create a **new source dataset without 0 to modify the relative url field**.

Create a parameter - *param_month_greaterthan9*. Then update the exp for relative url as below -
`trip-data/green_tripdata_2025-@{dataset().param_month_greaterthan9}.parquet`

Parameter value -> Update **`@item()`**

Pipeline sequence

For Each activity -> If Condition activity (inside ForEach) -> Copy activity (True) -> Copy activity (False)



Parameters Variables Settings Output

Activity name	Activity status	Activity ...	Run start	Duration
Copy data_greater_than_9	Succeeded	Copy data	6/15/2026, 10:52:47 AM	16s
If Condition	Succeeded	If Condition	6/15/2026, 10:52:46 AM	17s
Copy data_greater_than_9	Succeeded	Copy data	6/15/2026, 10:52:29 AM	16s
If Condition	Succeeded	If Condition	6/15/2026, 10:52:29 AM	18s
Copy data_greater_than_9	Succeeded	Copy data	6/15/2026, 10:52:12 AM	15s
If Condition	Succeeded	If Condition	6/15/2026, 10:52:12 AM	17s
Copy data less than 9	Succeeded	Copy data	6/15/2026, 10:51:56 AM	15s
If Condition	Succeeded	If Condition	6/15/2026, 10:51:55 AM	17s
Copy data less than 9	Succeeded	Copy data	6/15/2026, 10:51:39 AM	15s
If Condition	Succeeded	If Condition	6/15/2026, 10:51:38 AM	17s
Copy data less than 9	Succeeded	Copy data	6/15/2026, 10:51:21 AM	15s
If Condition	Succeeded	If Condition	6/15/2026, 10:51:20 AM	18s
Copy data less than 9	Succeeded	Copy data	6/15/2026, 10:51:02 AM	17s
If Condition	Succeeded	If Condition	6/15/2026, 10:51:01 AM	19s
Copy data less than 9	Succeeded	Copy data	6/15/2026, 10:50:44 AM	16s
If Condition	Succeeded	If Condition	6/15/2026, 10:50:43 AM	18s

Step 4 – Create Azure Databricks resource in MS Azure (SILVER LAYER)
Step 5 – Create compute in Databricks
Step 6 – Create Microsoft Entra Id (this will allow databricks to access the data in the data lake)
-> We will **create a service application** which will access data in the azure data lake.

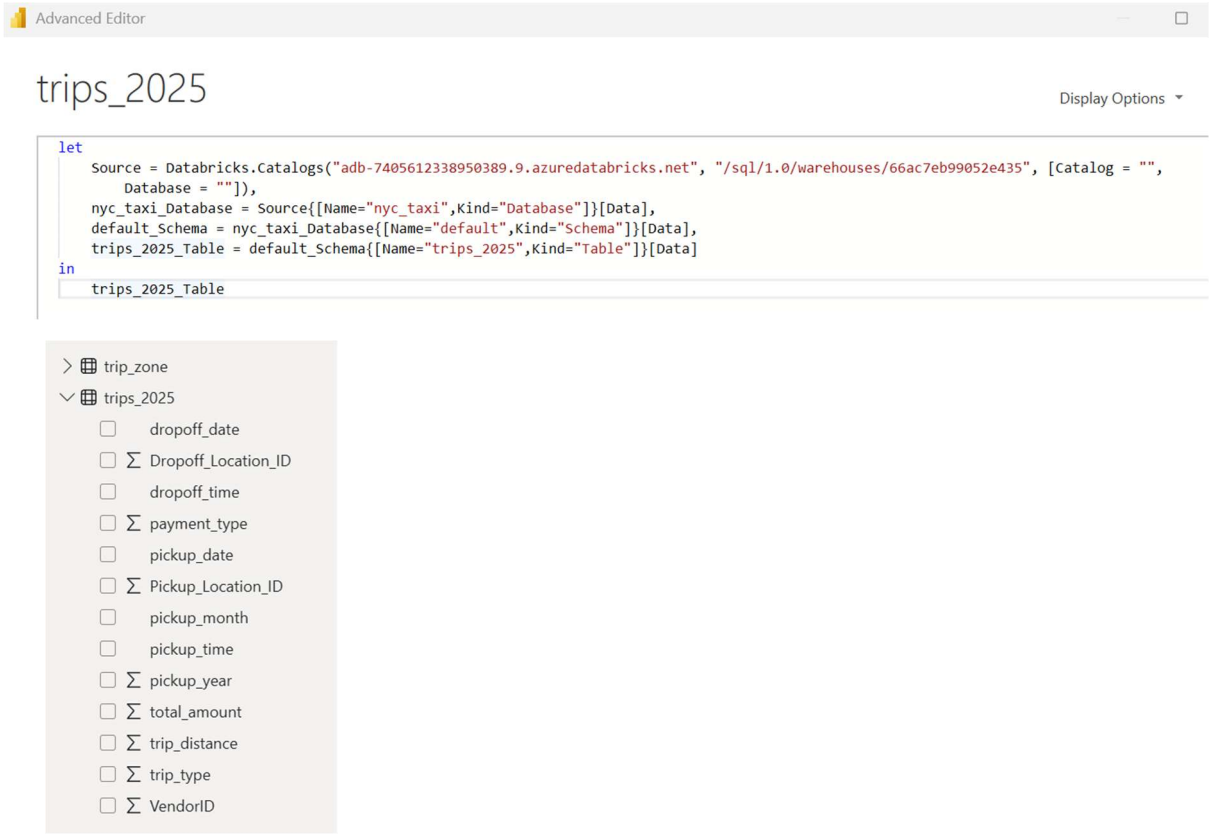
Step 7 – Launch Databricks Workspace
Go to Workspace and create a folder – *NYC_Project*
Create a new Notebook -> Name it as *Silver Layer Transformations*.

i. Run the below code to access data in the data lake -

```
spark.conf.set('fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net',
'OAuth')
spark.conf.set('fs.azure.account.oauth.provider.type.<storage-
account>.dfs.core.windows.net',
'org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider')
spark.conf.set('fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net',
'<application-id>')
spark.conf.set('fs.azure.account.oauth2.client.secret.<storage-
account>.dfs.core.windows.net', 'service credential')
spark.conf.set('fs.azure.account.oauth2.client.endpoint.<storage-
account>.dfs.core.windows.net', 'https://login.microsoftonline.com/<directory-
id>/oauth2/token')
```

iii. Perform transformations in the dataset.
iv. Write the data to MS Azure – **Silver** folder.

Step 8 - Connect to Power BI
-> Connect to Power BI using **Partner Connect.**


Advanced Editor
trips_2025
Display Options
let
Source = Databricks.Catalogs("adb-7405612338950389.9.azuredatabricks.net", "/sql/1.0/warehouses/66ac7eb99052e435", [Catalog = "",
Database = ""]),
nyc_taxi_Database = Source{[Name="nyc_taxi",Kind="Database"]}[Data],
default_Schema = nyc_taxi_Database{[Name="default",Kind="Schema"]}[Data],
trips_2025_Table = default_Schema{[Name="trips_2025",Kind="Table"]}[Data]
in
trips_2025_Table
trip_zone
trips_2025
dropoff_date
Dropoff_Location_ID
dropoff_time
payment_type
pickup_date
Pickup_Location_ID
pickup_month
pickup_time
pickup_year
total_amount
trip_distance
trip_type
VendorID